

How resilient is university mobility? Evaluating public transport disruptions with MATSim

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In person presentation preferred

1. Introduction

This research is part of the MovInSim project (Modeling resilient transportation systems under moderate flooding) funded by the Ministry of Science, Innovation and Universities. This abstract presents the preliminary results of this ongoing study.

The growing frequency of disruptions in urban transport systems (caused by extreme weather events, strikes, technical failures or energy crises among others) highlights the need to incorporate resilience as a strategic attribute in transport planning. In this context, Agent-Based Models (ABM) emerge as useful tools for evaluating user behaviour under adverse mobility scenarios, overcoming the limitations of classical aggregate models that fail to capture individual heterogeneity and the adaptive dynamics of travelers.

This work applies the Multi-Agent Transport Simulation (MATSim) framework to study the mobility of the Vera Campus of the Polytechnic University of Valencia (UPV), a trip-attracting institution hosting approximately 29,000 daily users, whose strong dependence on public transport makes it a particularly relevant case study for resilience analysis. The primary data source is the UPV Mobility Survey of May 2022, conducted as part of the UPV Strategic Sustainable Mobility Plan 2023–2027.

The scenario developed for this study relies on several simplifications, including a limited population consisting of the university community and a geographical scope restricted to the city of Valencia. The objective of this research is to develop and test a MATSim scenario within these constraints. As a newly initiated project, this work represents a significant and promising step forward in its overall development.

2. Methodology

The construction of the simulation scenario required the development and processing of five fundamental input files for MATSim: the Valencia road network (Network.xml), extracted and converted from OpenStreetMap using the JOSM editor; the agent activity plans (Plans.xml), derived from the statistical treatment of the mobility survey; and the public transport files (TransitSchedule.xml and TransitVehicles.xml), generated from the

GTFS public data from EMT and FGV operators, covering urban bus, metro and tram lines.

The simulated population consists of 1,971 agents, representing the university community, all residing within Valencia city, to whom a residential location was randomly assigned within georeferenced residential polygons at the neighbourhood level using QGIS. The initial modal choice for each agent was determined using the Monte Carlo method, calibrating mode-specific probabilities as a function of the distance to campus and the modal split reported in the official UPV survey. The transport modes considered were private car, public transport (PT), bicycle and walking.

Model calibration included a sensitivity analysis of road and public transport capacity across five levels (1%–5%), selecting the 2% scenario as the one that best balances inter-agent competition for network capacity and the necessary slack in the public transport system to detect the effect of disruptions.

The restrictive scenarios were structured in two blocks. The first involved the progressive and cumulative removal of the 7 public transport lines with direct access to campus (tram 6, tram 4, bus 98, bus 93, bus 71, bus 40 and bus 18), generating 6 scenarios ranging from full service to only two remaining lines. The second block evaluated system behaviour when only a single line is in operation, independently simulating each of the 7 lines as the sole available service. As a proposed solution, an active mobility incentive policy was formulated, reconfiguring 20% of public transport users towards bicycle use. A Python script was developed to modify the “Plans.xml” file.

3. Results

Results from the baseline scenario representing UPV mobility (Vera Campus) with full public transport service provided a modal split of 27.1% for bicycle, 25% for public transport, 17.5% for private car and 30.4% for walking, with a mean convergence utility of 118.87 points. The analysis of modal distribution as a function of distance showed consistent distribution with the literature: walking dominates the 0–1 km range, cycling between 1 and 5 km, and public transport and private car predominate for trips exceeding 5 km.

In the progressively restrictive scenarios, results confirm the redundancy effect of the system: the isolated removal of tram line 6 produced no significant variation in global utilities, as tram line 4 and bus 98 absorbed the displaced demand. Similar results were obtained removing line 98, at which the initial agent utility in the first iteration dropped from 46.3 to 18.3 points, evidencing a structural loss of north-south connectivity to campus. In the most restrictive scenario (with only lines 40 and 18 in operation) the initial utility reached a negative value for the first time, confirming the systemic impact of accumulated disruption.

In the single-line operation analysis, bus line 98 (EMT) and tram line 4 (FGV) resulted as the most critical axes for campus connectivity, reporting the highest convergence utilities (117.5 and 117.2 points respectively) and the smallest gaps between best and worst plan utilities at iteration 60 (7.0 and 14.0 points respectively). At the opposite

extreme, bus line 18 presented the largest divergence (20.35 points), indicating that its indirect routing forces agents to explore a greater number of alternatives without achieving stable convergence. These findings are aligned with the theoretical expectation that lines with lower spatial coverage of the simulated population generate greater variability in retained plans and wider utility gaps at convergence.

Promotion of active mobility significantly improved initial utility values both for gradual restriction scenarios and for single-line scenarios, compared to the baseline values without intervention. However, by iteration 60, the improvement in convergence utility was modest: +0.09% in best utilities and -0.57% in worst utilities for gradual restriction, and +0.19% and +3.04% respectively for single-line scenarios. This result reflects the distance-based disutility penalty experienced by agents reassigned from public transport to cycling, particularly those living far from campus. These results demonstrate that active mobility is a highly effective solution for the immediate relief of a disruption, but insufficient as a standalone response to structural disruptions affecting users with long commuting distances or with functional dependence on public transport.

4. Conclusions

This work confirms MATSim's potential as a public transport resilience assessment tool within the Spanish context, representing a novel empirical application for the university mobility domain in Valencia. The results allow the identification of the most critical transport lines for Vera campus connectivity (EMT bus line 98 and FGV tram line 4) the quantification of the non-linear impact of cumulative disruptions on user utility and the assessment of the real impact of active mobility incentive measures.

System resilience cannot rely on a single solution, it requires the development of mixed contingency plans that combine the deployment of active mobility options with reinforcement and substitution protocols for mass public transport, differentiating responses according to user profiles and the geographic extent of the disruption. In particular, the results highlight that a significant proportion of public transport users cannot be adequately served by active mobility alternatives alone, reinforcing the need for complementary emergency transit measures during service disruptions.

Future research lines include the extension of the model to the Valencia Metropolitan Area with a synthetic population of 10% of total inhabitants, the incorporation of mobile phone mobility data to improve spatial calibration, the application of the within-day replanning module to simulate intraday disruptions under partial information conditions, and the integration of econometrically calibrated discrete choice model coefficients to refine the agent scoring function and better represent heterogeneous preferences across different user groups.

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